P.I.R.E. INVENTORY RISK & CLEARANCE AUDIT REPORT

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1. Executive Portfolio Risk Summary

Total SKU Records Audited:	134	**Total Stock Value:**	■1,354,656.32
Expired Products (0% Sale):	**33**	**Inventory Value at Risk:**	**■1,035,105.88**
Critical & High Urgency:	43	**Est. Recoverable Value:**	**■498,598.44**
Medium & Low Risk:	58	**Expired Write-Off Loss:**	**■309,374.98**

■■ MANDATORY FOOD SAFETY DIRECTIVE: 33 product(s) have exceeded their expiry date. Retail food-safety regulations strictly forbid selling, remarketing, or discounting expired perishable items. These products must be quarantined immediately and removed for safe disposal.

2. Machine Learning (XGBoost) Model Validation

Classifier Architecture:	XGBoost Multi-Class Gradient Boosting	**Accuracy Score:**	100.0%
Weighted Precision:	100.0%	**Weighted Recall:**	100.0%
Weighted F1-Score:	100.0%	**Training Samples:**	107 train / 27 test

3. Priority Action Inventory Schedule

Key inventory records requiring immediate store manager attention:

PRD-1001 - Hummus Classic 200g	Packaged Foods	-8.0	**EXPIRED**	■176.8	Remove from Sale
PRD-1002 - Cold-Pressed Orange	Beverages	-2.0	**EXPIRED**	■100.2	Remove from Sale
PRD-1003 - Prawns Peeled 250g	Meat	-2.0	**EXPIRED**	■366.9	Remove from Sale
PRD-1004 - Fresh Pasta Tagliate	Packaged Foods	-3.0	**EXPIRED**	■156.9	Remove from Sale
PRD-1005 - Cheddar Cheese 200g	Dairy	-3.0	**EXPIRED**	■179.7	Remove from Sale
PRD-1006 - Fresh Cream 250ml	Dairy	-4.0	**EXPIRED**	■72.3	Remove from Sale
PRD-1007 - Salted Butter 500g	Dairy	-5.0	**EXPIRED**	■251.6	Remove from Sale

PRD-1008 - Chocolate Muffins (P	Bakery	-3.0	EXPIRED	■106.0	Remove from Sale
PRD-1009 - Fresh Cream 250ml	Dairy	-6.0	EXPIRED	■66.4	Remove from Sale
PRD-1010 - Probiotic Drink (5x6	Beverages	-3.0	EXPIRED	■75.6	Remove from Sale
PRD-1011 - Whole Wheat Bread 40	Bakery	-5.0	EXPIRED	■48.1	Remove from Sale
PRD-1012 - Ripe Strawberries 20	Fruits & Vegetables	-4.0	EXPIRED	■108.4	Remove from Sale
PRD-1013 - Cold Brew Coffee 250	Beverages	-3.0	EXPIRED	■114.7	Remove from Sale
PRD-1014 - Greek Yogurt 400g	Dairy	-6.0	EXPIRED	■83.1	Remove from Sale
PRD-1015 - Fresh Coconut Water	Beverages	-8.0	EXPIRED	■53.0	Remove from Sale
PRD-1016 - Whole Wheat Bread 40	Bakery	-4.0	EXPIRED	■49.4	Remove from Sale
PRD-1017 - Probiotic Drink (5x6	Beverages	-7.0	EXPIRED	■87.2	Remove from Sale
PRD-1018 - Prawns Peeled 250g	Meat	-8.0	EXPIRED	■372.9	Remove from Sale
PRD-1019 - Fresh Chicken Breast	Meat	0.0	EXPIRED	■218.9	Remove from Sale
PRD-1020 - Mutton Curry Cut 500	Meat	1.0	CRITICAL	■571.3	Discount 50.0% (■285.7)
PRD-1021 - Fresh Salsa Dip 200g	Packaged Foods	1.0	CRITICAL	■131.9	Discount 45.0% (■72.5)
PRD-1022 - Greek Yogurt 400g	Dairy	0.0	EXPIRED	■73.8	Remove from Sale
PRD-1023 - Salted Butter 500g	Dairy	0.0	EXPIRED	■256.5	Remove from Sale
PRD-1024 - Chocolate Muffins (P	Bakery	0.0	EXPIRED	■124.3	Remove from Sale
PRD-1025 - Bell Peppers Tri-col	Fruits & Vegetables	0.0	EXPIRED	■91.2	Remove from Sale

Audit Compliance Note: This report is generated by the P.I.R.E. Decision Support System. Model predictions are generated via an XGBoost multi-class classifier using temporal and cold-chain features. Estimated recoverable values represent potential revenue assuming 100% clearance of discount-eligible quantities and do not constitute guaranteed revenue. Perishable items past their stated expiration date are automatically condemned in strict alignment with retail food safety standards.